Theorem 6.3

The scalar triple product preserves addition and scalar multiplication. That is,

$$\begin{aligned}
[(\vec{a}+\vec{b}),\vec{c},\vec{d}] &= [\vec{a},\vec{c},\vec{d}]+[\vec{b},\vec{c},\vec{d}]\,;\\
[\lambda\vec{a},\vec{b},\vec{c}] &= \lambda[\vec{a},\vec{b},\vec{c}],\forall\lambda\in\mathbb{R}\\
[\vec{a},(\vec{b}+\vec{c}),\vec{d}] &= [\vec{a},\vec{b},\vec{d}]+[\vec{a},\vec{c},\vec{d}]\,;\\
[\vec{a},\lambda\vec{b},\vec{c}] &= \lambda[\vec{a},\vec{b},\vec{c}],\forall\lambda\in\mathbb{R}\\
[\vec{a},\vec{b},(\vec{c}+\vec{d})] &= [\vec{a},\vec{b},\vec{c}]+[\vec{a},\vec{b},\vec{d}]\,;\\
[\vec{a},\vec{b},\lambda\vec{c}] &= \lambda[\vec{a},\vec{b},\vec{c}],\forall\lambda\in\mathbb{R}\,.
\end{aligned}$$

Proof

Using the properties of scalar product and vector product, we get

$$\begin{aligned}
[(\vec{a}+\vec{b}),\vec{c},\vec{d}] &= ((\vec{a}+\vec{b})\times\vec{c})\cdot\vec{d}\\
&= (\vec{a}\times\vec{c}+\vec{b}\times\vec{c})\cdot\vec{d}\\
&= (\vec{a}\times\vec{c})\cdot\vec{d}+(\vec{b}\times\vec{c})\cdot\vec{d}\\
&= [\vec{a},\vec{c},\vec{d}]+[\vec{b},\vec{c},\vec{d}]\\
[\lambda\vec{a},\vec{b},\vec{c}] &= ((\lambda\vec{a})\times\vec{b})\cdot\vec{c}=(\lambda(\vec{a}\times\vec{b}))\cdot\vec{c}=\lambda((\vec{a}\times\vec{b})\cdot\vec{c})=\lambda[\vec{a},\vec{b},\vec{c}]\,.
\end{aligned}$$

Using the first statement of this result, we get the following.

$$\begin{aligned}
[\vec{a},(\vec{b}+\vec{c}),\vec{d}] &= [(\vec{b}+\vec{c}),\vec{d},\vec{a}]=[\vec{b},\vec{d},\vec{a}]+[\vec{c},\vec{d},\vec{a}]\\
&= [\vec{a},\vec{b},\vec{d}]+[\vec{a},\vec{c},\vec{d}]\\
[\vec{a},\lambda\vec{b},\vec{c}] &= [\lambda\vec{b},\vec{c},\vec{a}]=\lambda[\vec{b},\vec{c},\vec{a}]=\lambda[\vec{a},\vec{b},\vec{c}]\,.
\end{aligned}$$

Similarly, the remaining equalities are proved. ■

We have studied about coplanar vectors in XI standard as three nonzero vectors of which, one can be expressed as a linear combination of the other two. Now we use scalar triple product for the characterisation of coplanar vectors.

Theorem 6.4

The scalar triple product of three non-zero vectors is zero if, and only if, the three vectors are coplanar.

Proof

Let $\vec{a},\vec{b},\vec{c}$ be any three non-zero vectors. Then,

$$\begin{aligned}
(\vec{a}\times\vec{b})\cdot\vec{c}=0 &\Leftrightarrow \vec{c} \text{ is perpendicular to } \vec{a}\times\vec{b}\\
&\Leftrightarrow \vec{c} \text{ lies in the plane which is parallel to both } \vec{a} \text{ and } \vec{b}\\
&\Leftrightarrow \vec{a},\vec{b},\vec{c} \text{ are coplanar.}
\end{aligned}$$

■

Theorem 6.5

Three vectors $\vec{a},\vec{b},\vec{c}$ are coplanar if, and only if, there exist scalars $r,s,t\in\mathbb{R}$ such that atleast one of them is non-zero and $r\vec{a}+s\vec{b}+t\vec{c}=\vec{0}$.

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